

Marketing Mix Model: Bootstrap & Confidence Interval Statistics

Statistical Metrics Explained

- **rn**: Row number identifier for referencing specific observations
- **cluster**: Grouping of similar model solutions with comparable characteristics
- **n**: Sample size - number of observations used in the statistical analysis
- **boot_mean**: Average value calculated from bootstrap resampling process *Indicates the central tendency of parameter estimates across resamples*
- **boot_se**: Standard error from bootstrap analysis *Measures the precision of your parameter estimates*
- **boot_ci**: Width of the confidence interval *Shows the range of uncertainty in your estimates*
- **ci_low**: Lower bound of the confidence interval *Minimum likely value at your chosen confidence level*
- **ci_up**: Upper bound of the confidence interval *Maximum likely value at your chosen confidence level*
- **sd**: Standard deviation of the data *Measures the amount of variation or dispersion in your dataset*
- **dist100**: Distribution metric normalized to a 0-100 scale *Allows for standardized comparison between different variables*

Practical Interpretation

These bootstrap and confidence interval statistics help determine:

1. How reliable each channel's contribution estimate is
2. Whether a channel's effect is statistically significant
3. The potential range of ROI for each marketing investment
4. The consistency of model outcomes under different data conditions

Note: Wider confidence intervals (larger gap between ci_low and ci_up) indicate less certainty in the estimate.